

Supplementary Material

Lineage and Release Timing in Correlated Errors of Open-Weight Language Models: A Pair-Level Study

Sudharsan S, S. Kanaga Suba Raja, Shree Harish V,
Chin-Shiuh Shieh, Mong-Fong Horng, Lavanya R

This supplement holds material moved from the main paper: the per-model precision analysis that motivates the pair-level design (S1), the record of findings withdrawn from the earlier version of this work (S2), the simulation check and its post-hoc audit (S3), and detailed robustness results for every sample (S4). Every table and figure is generated from the committed outputs by the repository’s table and figure scripts.

S1. Why per-model variance decomposition is not enough

Model and identification

Let Y_i be a per-model trait (e.g., accuracy), with family $f(i)$ and release quarter $e(i)$:

$$\begin{aligned} Y_i &= \mu + \alpha_{f(i)} + \beta_{e(i)} + u_i, \\ \alpha &\sim \mathcal{N}(0, \sigma_L^2), \quad \beta \sim \mathcal{N}(0, \sigma_E^2), \quad u \sim \mathcal{N}(0, \sigma_U^2), \end{aligned} \tag{1}$$

so that $\text{Var}(\mathbf{y}) = \sigma_L^2 Z_F Z_F^\top + \sigma_E^2 Z_E Z_E^\top + \sigma_U^2 I$, with only the intercept as a fixed effect. The variance components are identified when the covariance basis $\{Z_F Z_F^\top, Z_E Z_E^\top, I\}$ is linearly independent. The rank of the fixed-effects dummy design $[\mathbf{1} \mid Z_F \mid Z_E]$, used as the gate in the earlier version of this work, is not the relevant condition for this model: for the sixteen models evaluated there, that design is rank-deficient (rank 14 of 15) while the covariance basis has full rank (3 of 3). Its condition number also depends on the arbitrary choice of reference level (TABLE SI).

Precision

Identification does not imply precision. For each design we compute the expected restricted-likelihood information $I_{jk} = \frac{1}{2} \text{tr}(PV_j PV_k)$ at four share scenarios (lineage-dominant 0.5/0.2/0.3, era-dominant 0.2/0.5/0.3, balanced, and low-signal 0.1/0.1/0.8), the delta-method standard error of each share, and, as the verdict, the Monte Carlo root-mean-square error (RMSE) of the share estimates under the same REML estimator (500 replicates per scenario). The analytic standard error tracks the Monte Carlo RMSE closely (e.g., 0.289 versus 0.285 for the sixteen-model design), so it can be used before any measurement.

Table SI: Precision of family and era variance shares for per-model traits. FE: fixed-effects dummy design used as a gate in the earlier version; κ range is over every choice of reference levels. Covariance-basis rank 3/3 means the variance components are identified. The last column is the worst-case Monte Carlo RMSE of either share across four scenarios (500 replicates each).

Design	Families	Quarters	FE full rank	$\kappa(X^\top X)$ range	Cov. basis	Worst RMSE
16 evaluated models	5	11	no	singular	3/3	0.302
22-model selection	6	14	yes	285–1526	3/3	0.256
47-model candidate frame	6	14	yes	166–1116	3/3	0.189
30-model “sufficient” design	6	8	yes	91–144	3/3	0.194
42-model sweep design	7	8	yes	83–113	3/3	0.182
nested reference (72)	6	6	no	singular	2/3	not identified

No design reaches an RMSE of 0.10 for both shares (TABLE SI): not the sixteen evaluated models (0.30), not the 47-model candidate frame (0.19), and not the 30-model configuration the earlier version described as sufficient (0.19). Removing the single model responsible for the fixed-effects rank deficiency leaves the RMSE unchanged (0.302). The binding constraint is the number of *levels* of each factor: more models per level help only up to a point (with six families and fourteen quarters, 22 models give 0.256 and 47 give 0.189). In staggered designs (Fig. S1), bringing both shares within ± 0.10 requires on the order of sixty families and fourteen quarters; with eight quarters the era share never falls below about 0.14. Because release quarters are few by construction, a per-model decomposition is a poor instrument for the paper’s research questions in populations like those examined here. Within the designs and scenarios examined, this motivates the pair-level design; it does not show that the shares are inestimable under every design, only that the required numbers of families and quarters exceed those available.

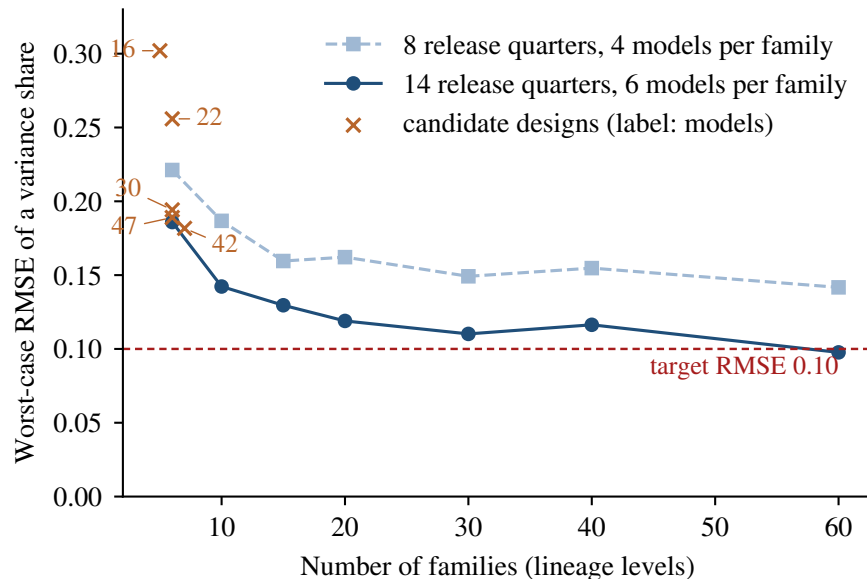


Figure S1: Precision of per-model variance shares versus the number of families. Worst-case Monte Carlo RMSE of the family or era share across four scenarios (300 replicates each) for staggered designs with 8 or 14 release quarters, and for the five candidate designs of TABLE SI (crosses, labelled by number of models; 5–7 families). The RMSE is a Monte Carlo root-mean-square error of the estimated share, not the half-width of a confidence interval. The dashed line marks 0.10; only the largest staggered design with 14 quarters (60 families, 360 models) reaches it.

S2. Changes from the earlier version

The earlier version of this manuscript (archived in the repository as [paper/src/archive/](#)) reported a sixteen-model evaluation and an identifiability gate. TABLE SII lists each finding of that version that is withdrawn or corrected, why, and how it is treated here.

Table SII: Findings of the earlier version and their treatment.

Earlier claim	Reason	Current treatment
Variance components “not identifiable” (fixed-effects dummy-design gate)	Random-effects model identified (basis rank 3/3); condition number depends on reference level (115–585)	Withdrawn; imprecision quantified (Section S1)
Item-level predictions; Qwen-7B accuracy 0.2294	Code read only option A’s log-likelihood; 0.2294 is the all-A rate	Withdrawn; code fixed; validation against stored correctness (main paper, Section III)
30-model design “passes” at $\kappa \leq 50$	Its κ is 93	Corrected
Sweep enforced crossing and connectedness	100 of 105 crossed, 60 connected	Corrected
LPM and GLMM both hit the era boundary in 20–60% of replicates	Only the GLMM; LPM 0–6.7%	Corrected
Nested-design detector: “zero silent coverage”	Undefined: no replicate went undetected	Corrected
Sixteen-model population as pre-specified	Four planned models not evaluated, no recorded reason; two DeepSeek models imputed	Arm not used for inference

S3. Simulation check of separability

The main paper (Section IV-E) describes the simulated data and criteria. Table SIII gives the pre-specified check (20 replicates per scenario) for three caps on the number of models per root; Table SIV and Fig. S2 give the post-hoc audit with 500 replicates. With 20 replicates the rates move in steps of five points, which explains why the audit ranks the caps differently. Cap 40 remains the primary design because it was the dated choice made before any pair outcome was computed.

Table SIII: Pre-specified simulation check (primary roster, 14,042 simulated items, 20 replicates per scenario). Null: largest false-rejection rate. Leakage: largest rejection of the wrong coefficient. Power: rejection of the lineage / same-month coefficient at the stated effect weight.

Cap	N	Null	Leakage	Power $\lambda=0.05$	Power $\lambda=0.03$
15	561	15%	20%	100% / 100%	50% / 75%
40	613	5%	10%	100% / 100%	40% / 95%
none	721	10%	15%	100% / 100%	95% / 95%

Table SIV: Post-hoc gate audit (500 replicates per truth; Wilson 95% intervals, in percent). Null: largest rejection rate of any term under no effect. Leakage: largest cross-term rejection across $\lambda \in \{0.03, 0.05, 0.10\}$. Power: rejection of the lineage (L) or release-month (E) term. Post hoc.

Sample	Cap	N	Null	Leakage	L, $\lambda=0.05$	E, $\lambda=0.05$	L, $\lambda=0.03$	E, $\lambda=0.03$
Primary	15	561	6.0 [4.2, 8.4]	7.2 [5.2, 9.8]	99.6 [98.6, 99.9]	100.0 [99.2, 100.0]	40.2 [36.0, 44.6]	83.8 [80.3, 86.8]
Primary	40	613	7.8 [5.8, 10.5]	10.8 [8.4, 13.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	63.2 [58.9, 67.3]	87.8 [84.6, 90.4]
Primary	none	721	6.6 [4.7, 9.1]	7.2 [5.2, 9.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	90.4 [87.5, 92.7]	90.2 [87.3, 92.5]
Expanded	40	967	5.6 [3.9, 8.0]	12.6 [10.0, 15.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	93.6 [91.1, 95.4]	97.8 [96.1, 98.8]

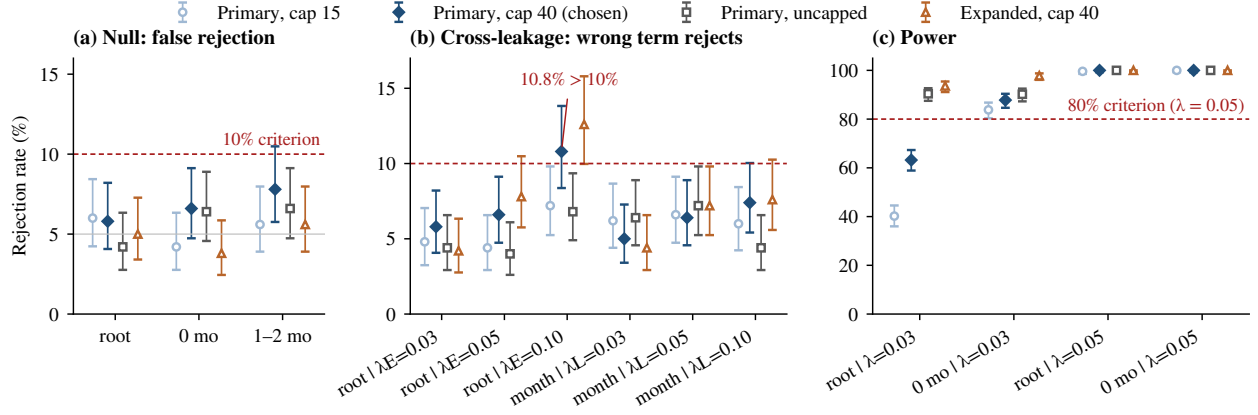


Figure S2: Post-hoc audit of the simulation check (500 replicates per scenario; Wilson 95% intervals). (a) False rejection of each coefficient under no effect. (b) Leakage: rejection of the lineage coefficient under release-time-only effects (λ_E), and the larger rejection rate of the 0-month and 1–2-month coefficients under lineage-only effects (λ_L). (c) Power. Dashed lines: the pre-specified criteria. The primary cap-40 design exceeds the leakage criterion at $\lambda_E = 0.10$ (10.8%, interval 8.4–13.8%), as does the expanded design (12.6%).

S4. Detailed robustness results

The main paper’s robustness table reports the primary sample. The tables below give every sample; figures show the same estimates. “Same month” is the 0-month gap bin relative to 12 or more months. S2: models with accuracy ≥ 0.30 .

Table SV: Coefficients with and without the accuracy terms (95% CIs from the delete-one-root jackknife).

Term	Primary		Expanded	
	Without	With	Without	With
Shared root	+0.283 [+0.251, +0.315]	+0.133 [+0.108, +0.158]	+0.240 [+0.205, +0.275]	+0.127 [+0.110, +0.145]
Same month	+0.087 [+0.051, +0.122]	+0.004 [−0.005, +0.013]	+0.092 [+0.065, +0.119]	+0.006 [−0.002, +0.014]
1–2 months	+0.085 [+0.048, +0.123]	0.000 [−0.007, +0.007]	+0.087 [+0.060, +0.114]	+0.002 [−0.005, +0.010]
3–5 months	+0.065 [+0.031, +0.099]	−0.001 [−0.009, +0.007]	+0.078 [+0.050, +0.107]	+0.001 [−0.007, +0.008]
6–11 months	+0.029 [+0.012, +0.046]	−0.001 [−0.008, +0.006]	+0.055 [+0.031, +0.079]	+0.002 [−0.006, +0.009]

Table SVI: Inference procedures. [†]Pre-specified. Root-level procedures treat lineage roots as the unit of dependence.

Sample	Term	Standard error	Estimate [95% CI]	<i>p</i>
Primary	same root	model-dyadic [†]	+0.133 [+0.118, +0.148]	<0.001
Primary	same root	root-dyadic	+0.133 [+0.114, +0.152]	<0.001
Primary	same root	delete-one-root jackknife	+0.133 [+0.108, +0.158]	<0.001
Primary	same month	model-dyadic [†]	+0.0039 [−0.0029, +0.0106]	0.263
Primary	same month	root-dyadic	+0.0039 [−0.0041, +0.0118]	0.343
Primary	same month	delete-one-root jackknife	+0.0039 [−0.0048, +0.0125]	0.383
Primary, S2	same root	model-dyadic [†]	+0.120 [+0.105, +0.135]	<0.001
Primary, S2	same root	root-dyadic	+0.120 [+0.103, +0.136]	<0.001
Primary, S2	same root	delete-one-root jackknife	+0.120 [+0.098, +0.142]	<0.001
Primary, S2	same month	model-dyadic [†]	−0.0145 [−0.0268, −0.0022]	0.021
Primary, S2	same month	root-dyadic	−0.0145 [−0.0269, −0.0021]	0.022
Primary, S2	same month	delete-one-root jackknife	−0.0145 [−0.0303, +0.0013]	0.072
Expanded	same root	model-dyadic [†]	+0.127 [+0.115, +0.140]	<0.001
Expanded	same root	root-dyadic	+0.127 [+0.112, +0.143]	<0.001
Expanded	same root	delete-one-root jackknife	+0.127 [+0.110, +0.145]	<0.001
Expanded	same month	model-dyadic [†]	+0.0059 [−0.0003, +0.0121]	0.061
Expanded	same month	root-dyadic	+0.0059 [−0.0017, +0.0135]	0.128
Expanded	same month	delete-one-root jackknife	+0.0059 [−0.0023, +0.0142]	0.160
Expanded, S2	same root	model-dyadic [†]	+0.122 [+0.111, +0.133]	<0.001
Expanded, S2	same root	root-dyadic	+0.122 [+0.108, +0.136]	<0.001
Expanded, S2	same root	delete-one-root jackknife	+0.122 [+0.105, +0.139]	<0.001
Expanded, S2	same month	model-dyadic [†]	−0.0089 [−0.0185, +0.0008]	0.071
Expanded, S2	same month	root-dyadic	−0.0089 [−0.0191, +0.0013]	0.088
Expanded, S2	same month	delete-one-root jackknife	−0.0089 [−0.0218, +0.0040]	0.178

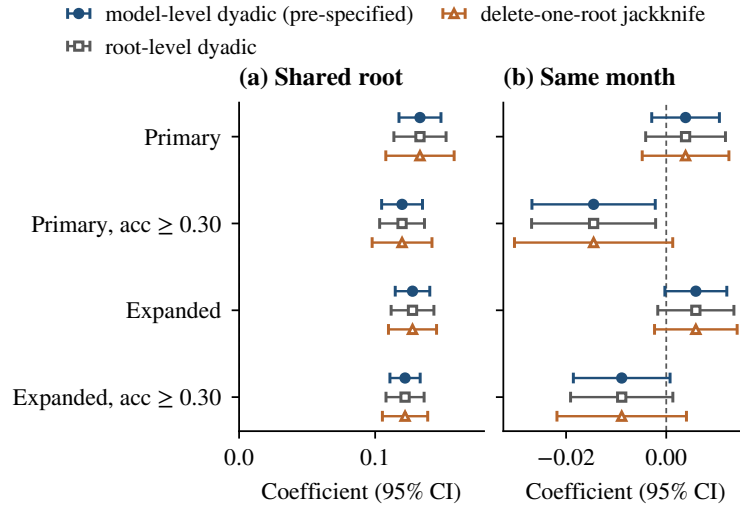


Figure S3: Inference procedures (95% CIs).

Table SVII: Outcome and weighting variants (95% CIs, model-level dyadic SEs). Every pair has at least 1,166 jointly wrong items, so the variant restricted to pairs with at least 1,000 is identical to the pre-specified fit.

Sample	Variant	Shared root	Same month	p (same month)
Primary	primary outcome, OLS (pre-specified)	+0.133 [+0.118, +0.148]	+0.0039 [-0.0029, +0.0106]	0.263
Primary	WLS, weight = jointly-wrong items	+0.142 [+0.125, +0.159]	+0.0117 [+0.0016, +0.0217]	0.023
Primary	pairs with ≥ 1000 jointly-wrong items	+0.133 [+0.118, +0.148]	+0.0039 [-0.0029, +0.0106]	0.263
Primary	phi of error indicators (secondary)	+0.170 [+0.149, +0.191]	+0.0184 [+0.0097, +0.0272]	<0.001
Primary	agreement minus option-distribution chance (exploratory)	+0.125 [+0.111, +0.140]	-0.0018 [-0.0046, +0.0011]	0.227
Expanded	primary outcome, OLS (pre-specified)	+0.127 [+0.115, +0.140]	+0.0059 [-0.0003, +0.0121]	0.061
Expanded	WLS, weight = jointly-wrong items	+0.136 [+0.122, +0.151]	+0.0114 [+0.0033, +0.0195]	0.006
Expanded	pairs with ≥ 1000 jointly-wrong items	+0.127 [+0.115, +0.140]	+0.0059 [-0.0003, +0.0121]	0.061
Expanded	phi of error indicators (secondary)	+0.163 [+0.146, +0.180]	+0.0194 [+0.0118, +0.0270]	<0.001
Expanded	agreement minus option-distribution chance (exploratory)	+0.116 [+0.105, +0.127]	+0.0007 [-0.0019, +0.0033]	0.604

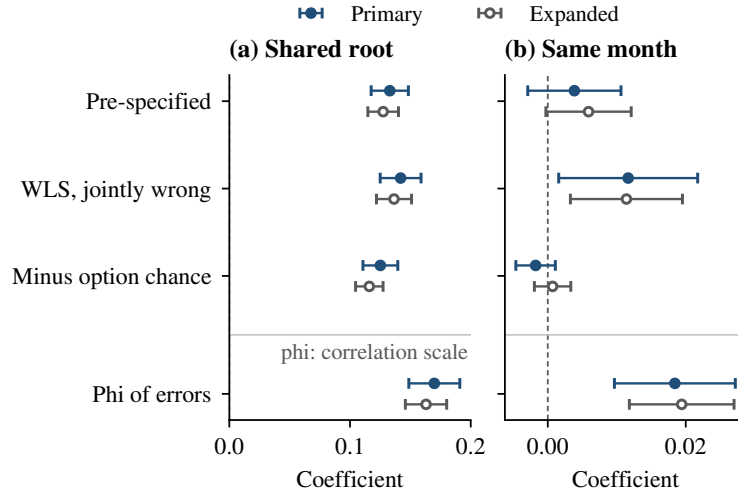


Figure S4: Outcome and weighting variants (95% CIs). Phi is a correlation, on a different scale.

Table SVIII: Item-difficulty-aware nulls (post hoc). Mean: average outcome across pairs. 95% CIs and p -values from the delete-one-root jackknife. N1a: observed agreement minus the item-level distractor null computed from all models; N1b: the same with roots weighted equally; N2: co-failure beyond a Rasch model, per item.

Sample	Outcome	Mean	Shared root	p	Same month	p
Primary	Same-wrong agreement (primary outcome)	0.423	+0.133 [+0.108, +0.158]	<0.001	+0.0039 [-0.0048, +0.0125]	0.383
Primary	N1a: minus item distractor null	0.002	+0.133 [+0.107, +0.159]	<0.001	+0.0040 [-0.0047, +0.0127]	0.367
Primary	N1b: as N1a, roots weighted equally	0.019	+0.133 [+0.108, +0.159]	<0.001	+0.0038 [-0.0049, +0.0124]	0.393
Primary	N2: co-failure over Rasch null (per item)	0.000	+0.040 [+0.032, +0.048]	<0.001	+0.0036 [+0.0012, +0.0060]	0.003
Primary, S2	Same-wrong agreement (primary outcome)	0.551	+0.120 [+0.098, +0.142]	<0.001	-0.0145 [-0.0303, +0.0013]	0.072
Primary, S2	N1a: minus item distractor null	0.000	+0.119 [+0.096, +0.142]	<0.001	-0.0138 [-0.0310, +0.0034]	0.116
Primary, S2	N1b: as N1a, roots weighted equally	0.023	+0.120 [+0.097, +0.143]	<0.001	-0.0143 [-0.0315, +0.0030]	0.104
Primary, S2	N2: co-failure over Rasch null (per item)	0.000	+0.035 [+0.028, +0.042]	<0.001	+0.0008 [-0.0019, +0.0036]	0.561
Expanded	Same-wrong agreement (primary outcome)	0.434	+0.127 [+0.110, +0.145]	<0.001	+0.0059 [-0.0023, +0.0142]	0.160
Expanded	N1a: minus item distractor null	0.004	+0.128 [+0.110, +0.146]	<0.001	+0.0061 [-0.0024, +0.0146]	0.160
Expanded	N1b: as N1a, roots weighted equally	0.024	+0.128 [+0.110, +0.146]	<0.001	+0.0062 [-0.0023, +0.0146]	0.154
Expanded	N2: co-failure over Rasch null (per item)	0.000	+0.038 [+0.032, +0.045]	<0.001	+0.0036 [+0.0012, +0.0060]	0.003
Expanded, S2	Same-wrong agreement (primary outcome)	0.547	+0.122 [+0.105, +0.139]	<0.001	-0.0089 [-0.0218, +0.0040]	0.178
Expanded, S2	N1a: minus item distractor null	0.000	+0.122 [+0.105, +0.140]	<0.001	-0.0088 [-0.0226, +0.0051]	0.214
Expanded, S2	N1b: as N1a, roots weighted equally	0.019	+0.123 [+0.106, +0.141]	<0.001	-0.0088 [-0.0229, +0.0052]	0.216
Expanded, S2	N2: co-failure over Rasch null (per item)	0.000	+0.037 [+0.031, +0.043]	<0.001	+0.0016 [-0.0028, +0.0060]	0.480

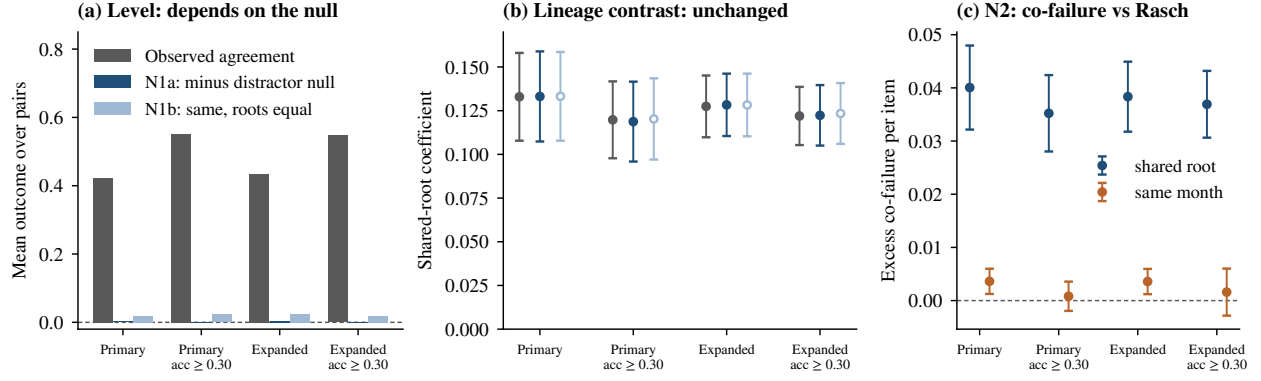


Figure S5: Item-difficulty-aware nulls. (a) Mean outcome over pairs: the level of agreement depends on the null. (b) Shared-root coefficient: the lineage contrast does not. (c) Co-failure beyond a Rasch model.

Table SIX: Size and architecture controls (exploratory; all include the S1 covariate). Entries: estimate (SE).

Sample	Models	Shared root (root-clustered SE)	Shared root (+ size, architecture)	Same month (+ size, architecture)	Same architecture
Primary	590	+0.1328 (0.0096)	+0.1256 (0.0082)	+0.0071 (0.0050)	+0.0027 (0.0026)
Primary, S2	343	+0.1166 (0.0086)	+0.1018 (0.0087)	-0.0205 (0.0110)	+0.0182 (0.0042)
Expanded	928	+0.1263 (0.0072)	+0.1201 (0.0062)	+0.0115 (0.0045)	+0.0050 (0.0022)
Expanded, S2	589	+0.1200 (0.0069)	+0.1085 (0.0057)	-0.0088 (0.0081)	+0.0181 (0.0024)

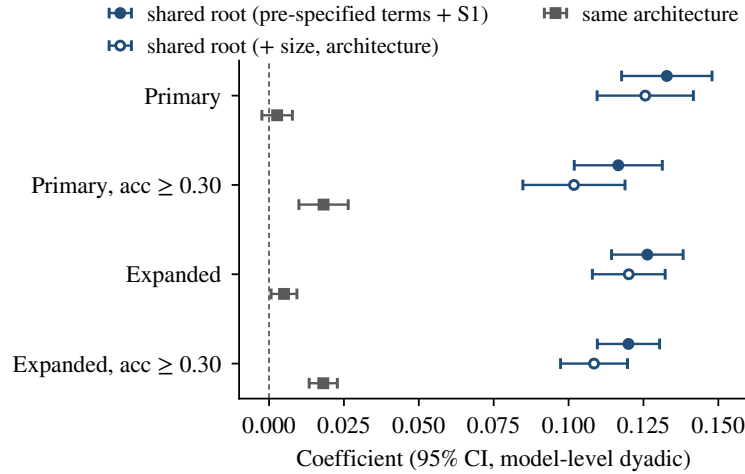


Figure S6: Size and architecture controls (95% CIs, model-level dyadic SEs).

Table SX: Further post-hoc analyses (delete-one-root jackknife 95% CIs). Equivalence bound: largest absolute same-month coefficient not excluded by the 90% interval. Root release month: gap bins from the roots' release months. CAPA: chance-adjusted agreement from chosen options. Clean sample: degenerate models removed and identical-prediction clusters collapsed. Config-consistent: models whose configuration file points to another root removed. Tree distance: primary sample only.

	Primary	Primary, S2	Expanded	Expanded, S2
Same month: 90% equivalence bound	± 0.011	± 0.028	± 0.013	± 0.020
Root release month: shared root	+0.131 [+0.108, +0.154]	+0.122 [+0.103, +0.140]	+0.119 [+0.101, +0.136]	+0.112 [+0.096, +0.127]
Root release month: same month	-0.001 [-0.011, +0.010]	-0.022 [-0.040, -0.004]	+0.009 [-0.004, +0.022]	+0.001 [-0.017, +0.019]
CAPA outcome: shared root	+0.194 [+0.153, +0.234]	+0.166 [+0.131, +0.201]	+0.183 [+0.156, +0.210]	+0.169 [+0.144, +0.195]
CAPA outcome: same month	+0.013 [+0.002, +0.023]	-0.002 [-0.013, +0.010]	+0.014 [+0.004, +0.023]	+0.002 [-0.010, +0.014]
Clean sample: shared root	+0.128 [+0.106, +0.150]	+0.117 [+0.096, +0.137]	+0.125 [+0.107, +0.142]	+0.120 [+0.104, +0.135]
Clean sample: same month	-0.002 [-0.008, +0.003]	-0.015 [-0.031, +0.001]	+0.002 [-0.005, +0.009]	-0.009 [-0.022, +0.004]
Config-consistent lineage: shared root	+0.135 [+0.108, +0.161]	-	-	-
Tree distance 1	+0.185 [+0.157, +0.213]	-	-	-
Tree distance 2	+0.136 [+0.117, +0.155]	-	-	-
Tree distance 3+	+0.115 [+0.103, +0.128]	-	-	-